

Latent Iteration as Renormalization

Inference-time recurrence in a depth-recurrent transformer flows attention geometry toward the SYK conformal fixed point, and the flow requires trained weights

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Abstract

Depth-recurrent transformers scale test-time compute by iterating a core block in latent space before emitting tokens. The geometry of these latent trajectories is a named open problem in the latent reasoning literature. Here we connect it to an independently developed body of measurements: trained transformer attention exhibits per-head power-law decay $A(i, j) \sim |i - j|^{-2\Delta}$ whose median exponent in deep layers matches the Sachdev–Ye–Kitaev (SYK) conformal value $\Delta = 1/4$, and prior work found the exponent flows toward that value along two depth axes — architectural layer depth and training time. We test the third axis: *inference-time* depth on fixed weights. In a pre-registered experiment on Huginn-0125, a 3.5B-parameter depth-recurrent transformer, we measure the per-head attention exponent of the recurrent core at recurrence counts $r \in \{1, 2, 4, 8, 16, 32\}$. For natural-text inputs the median exponent decreases monotonically with recurrence (Spearman $\rho = -0.94$), from $\Delta_{\text{med}} = 0.29$ at $r=1$ to 0.239 at $r=32$ — converging onto the SYK prediction — while the number of heads in the SYK-near window ($0.20 \leq \Delta \leq 0.30$, $R^2 > 0.90$) grows monotonically ($\rho = +0.77$). Random-token inputs show an architecture-driven convergence with a transient at $\Delta_{\text{med}} = 0.52 \approx 1/2$ at $r=2$ — the $q=2$ prethermal plateau previously observed in training time — but no monotone SYK-near growth ($\rho = +0.14$). A randomized-weights control freezes completely: Δ_{med} constant to the sixth decimal across all r , with zero SYK-near heads. Inference-time recurrence therefore acts as renormalization-group flow toward the conformal fixed point, and the flow is a property of the trained model, not of the iteration procedure. This gives the published orbit/spiral phenomenology of recurrent-depth latent trajectories a candidate theory — the spiral is flow near an infrared fixed point — and predicts that the geometric benefit of additional recurrence saturates when the flow arrives, consistent with the empirical success of convergence-based early-exit criteria. All code, pre-registration, and per-head data are in the public repository `3ld0n/attention-geometry`.

1. Introduction

A growing research program moves transformer computation out of explicit tokens and into continuous hidden states: latent chain of thought, latent memory and planning, and — most directly relevant here — *depth-recurrent* architectures that iterate a core block in latent space before emitting each token, making test-time compute a continuous dial (Geiping et al. 2025; Yu et al. 2026). The empirical phenomenology of these latent trajectories is already partially charted. Geiping et al. report structured PCA trajectories — orbits and spirals — in the latent iterates of their recurrent-depth model (Geiping et al. 2025); Pappone et al. find a two-scale picture in which loop updates shrink and become increasingly orthogonal, “tracing a stable-curvature spiral around a nearby representation,” and build a practical early-exit criterion on when that spiral stabilizes

(Pappone et al. 2025). What is missing is a theory of the geometry. The field’s own survey names “a deeper account of the geometry of latent trajectories” as a top open theoretical need (Yu et al. 2026, sec. 6.3).

Independently, a body of measurements has accumulated on the *attention geometry* of trained transformers. Per-head attention weight decays as a power law in query–key separation, $A(i, j) \sim |i - j|^{-2\Delta}$, and the median fitted exponent across power-law heads in GPT-2 is $\Delta = 0.2493$ — matching the conformal dimension $\Delta = 1/4$ of the Sachdev–Ye–Kitaev model at $q = 4$ (Umphrey and Umphrey 2026b; Sachdev and Ye 1993; Kitaev 2015; Maldacena and Stanford 2016). The exponent behaves like an infrared fixed point under two distinct notions of depth:

1. **Architectural depth.** Within a single trained model, the per-head exponent flows from $\bar{\Delta} \approx 0.70$ in early layers to $\bar{\Delta} = 0.250$ in deep layers; across separately trained models of increasing depth, the median exponent converges toward 0.25 (Umphrey and Umphrey 2026b, 2026a).
2. **Training time.** Along the training trajectory of Pythia-70m, the count of SYK-near heads grows monotonically with training step (Spearman $\rho = 0.86$ for random-token probes, 0.73 for natural text), passing through a transient plateau at $\Delta_{\text{med}} \approx 0.5$ — the free ($q=2$) value — before approaching the interacting $q=4$ value (repository experiment exp-086).

Both axes confound depth with other variables: architectural depth compares different parameters, and training time changes the weights at every step. A depth-recurrent model closes the gap. Recurrence count r is a *continuously adjustable depth variable on a single fixed set of trained weights*: if the conformal fixed point is a genuine attractor of iterative attending — rather than an artifact of feedforward training — the flow should appear at inference time, with no weight changes at all.

This paper reports a pre-registered test of that prediction on Huginn-0125 (Geiping et al. 2025), with two input conditions (natural text and random tokens) and a randomized-weights control. The result, stated at its measured strength: inference-time recurrence flows the attention exponent monotonically onto the SYK value for natural text; random-token inputs reveal an architecture-driven component of the convergence plus the same $q=2$ transient seen in training time; and randomizing the weights abolishes the flow entirely. Recurrence is renormalization — but only over a medium that training has prepared.

2. Background: the conformal fixed point in attention

We summarize only what is needed to read the measurement; full development is in the cited preprints.

For each attention head, average attention weight as a function of token separation $\Delta x = i - j$ defines a lag profile $A(\Delta x)$. In trained models a subpopulation of heads follows $A(\Delta x) \propto |\Delta x|^{-2\Delta}$ over an extended range. We fit Δ per head by log–log regression (Section 3) and call a measurement **SYK-near** when $0.20 \leq \Delta \leq 0.30$ with $R^2 > 0.90$. The reference value $\Delta = 1/4$ is the conformal dimension of the SYK model with four-fermion ($q=4$) interactions in the infrared, conformal limit; the free-fermion ($q=2$) value is $\Delta = 1/2$ (Maldacena and Stanford 2016). In renormalization-group (RG) language, the prior findings are that attention geometry flows from an ultraviolet regime (steep, large- Δ profiles) toward the $q=4$ infrared fixed point as depth increases, occasionally lingering at a $q=2$ prethermal plateau en route — a two-stage flow directly measured in training time (exp-086).

Two head populations matter for interpreting input conditions (repository experiments exp-

086/087/088). **Structural** conformal heads sit near the attractor by virtue of architecture and weight geometry; they respond to random-token inputs and appear early (in training time) or at low depth. **Semantic** conformal heads require natural-language input and long-range causal tracking; they form later and deeper. Random-token probes therefore isolate the structural component; natural text engages both.

3. Methods

Model. Huginn-0125 (tomg-group-umd/huginn-0125): 3.5B parameters, trained on 800B tokens (Geiping et al. 2025). Architecture: a 2-block prelude, a 4-block recurrent core executed r times per forward pass, and a 2-block coda; 55 attention heads per core layer, $d_{\text{head}} = 96$. Loaded in `bf16` with the model’s own remote code.

Probe. Recurrence counts $r \in \{1, 2, 4, 8, 16, 32\}$, passed as a fixed integer per forward pass. Inputs: 20 sequences of 128 tokens of natural text (TinyStories, the same source as our training-time experiments) and 20 sequences of uniform-random token IDs. For each r , each core layer, and each head, we capture the attention pattern at every recurrent step via forward hooks on the four core `CausalSelfAttention` modules, computing per-head attention manually from the module’s own weights — including Huginn’s native rotary embedding function and its exact query–key normalization and bias ordering, imported from the model’s own code rather than re-implemented. Total: 52,800 per-head measurements (20 sequences $\times$ 6 recurrence counts $\times$ 4 layers $\times$ 55 heads $\times$ 2 conditions).

Fitting. Per measurement, log–log regression of $A(\Delta x)$ against $\Delta x \in [3, 60]$; record (Δ, R^2) . Per recurrence step and condition we report the median exponent Δ_{med} over all 4400 (layer, head, sequence) measurements and the SYK-near count as defined above.

Pre-registered hypotheses (committed before model download, repository commit 702efd95):

- **H_emergence:** SYK-near count increases monotonically with r for natural text; criterion $\rho(r, \text{count}) > 0.5$.
- **H_convergence:** Δ_{med} decreases monotonically with r for natural text; criterion $\rho(r, \Delta_{\text{med}}) < -0.5$.
- **H_rand_contrast:** both correlations weaker in magnitude for random tokens than for natural text.
- **Kill criterion:** the result is not confirmed if $\rho(r, \text{count}) < 0.3$ and $\rho(r, \Delta_{\text{med}}) > -0.3$ for natural text.

Control. After the primary analysis, all weight tensors are randomized in place (i.i.d. normal, matched per-tensor standard deviation) and the natural-text condition is re-run identically. The pre-registered expectation is the absence of any systematic pattern.

Protocol deviations, recorded in the experiment notes. (1) The pre-registration specified local Apple-silicon execution; the 16GB download was infeasible on our rural connection and the run moved to a cloud A100-40GB (CUDA). (2) Two implementation errors were found and fixed before any results were produced: the recurrence count must be passed as a plain integer (the model’s schedule sampler interprets tensors differently), and the initial re-implementation of the rotary embedding assumed a complex-polar `freqs_cis` format while Huginn stores a cos/sin stack; the fix imports Huginn’s own `apply_rotary_emb_complex_like`. Both fixes increase fidelity to the model’s native forward pass; neither touches the fitting pipeline.

4. Results

4.1 Natural text: monotone flow onto the fixed point

Table 1: Natural-text condition. $\rho(r, \Delta_{\text{med}}) = -0.943$ (**H_convergence: keep**); $\rho(r, \text{SYK-near}) = +0.771$ (**H_emergence: keep**).

r	Δ_{med}	SYK-near (of 4400)
1	0.2916	4
2	0.3082	0
4	0.2761	2
8	0.2424	20
16	0.2392	26
32	0.2387	23

The median exponent decreases monotonically from 0.292 to 0.239, crossing into the SYK-near window at $r = 8$ and settling just below the $q=4$ value 0.25. SYK-near head formation is essentially absent through $r = 4$ and then rises by an order of magnitude. The flow **saturates at** $r \approx 8\text{--}16$: further recurrence does not move Δ_{med} below ≈ 0.239 . The fixed point behaves like a fixed point.

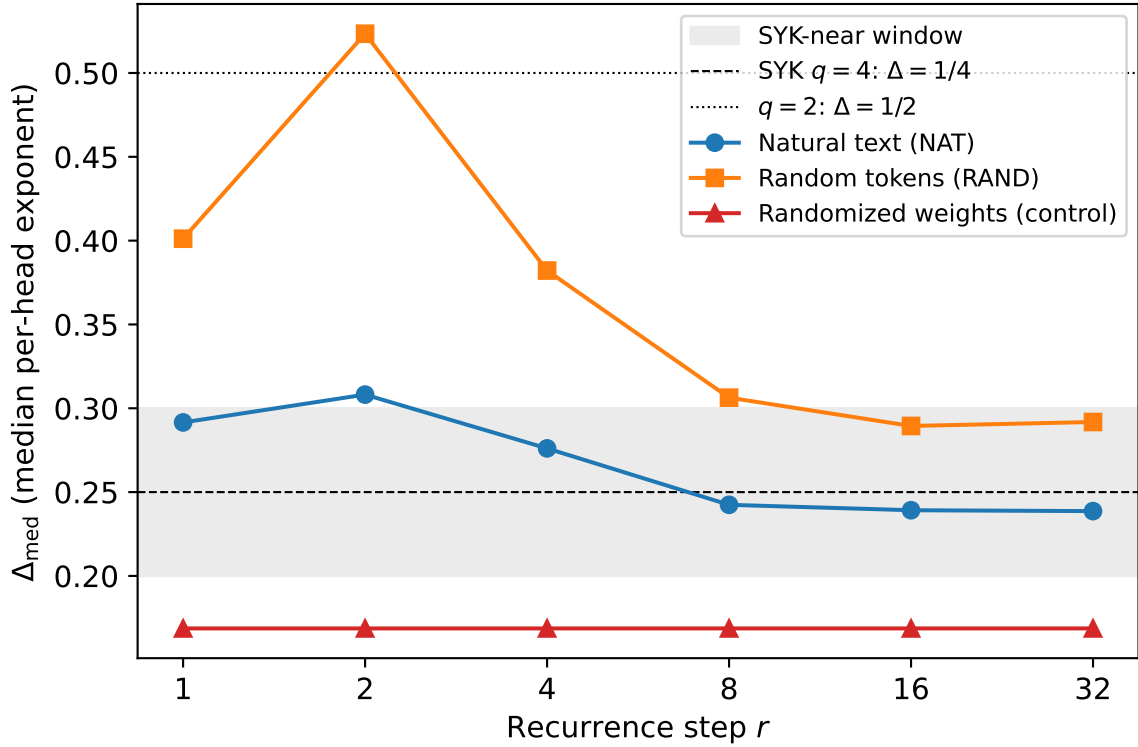


Figure 1: Median per-head exponent Δ_{med} versus recurrence step, for natural text, random tokens, and the randomized-weights control. Dashed line: SYK $q=4$ value $\Delta = 1/4$; dotted line: free $q=2$ value $\Delta = 1/2$; band: SYK-near window $[0.20, 0.30]$.

4.2 Random tokens: architectural component and the $q=2$ transient

Table 2: Random-token condition. $\rho(r, \Delta_{\text{med}}) = -0.886$; $\rho(r, \text{SYK-near}) = +0.143$.

r	Δ_{med}	SYK-near (of 4400)
1	0.4012	145
2	0.5234	48
4	0.3822	52
8	0.3063	91
16	0.2894	98
32	0.2918	99

Both contrasts pre-registered under **H_rand_contrast** hold: $|\rho_{\text{conv}}^{\text{NAT}}| > |\rho_{\text{conv}}^{\text{RAND}}|$ and $\rho_{\text{emerg}}^{\text{NAT}} > \rho_{\text{emerg}}^{\text{RAND}}$ (**keep**). Three features are informative beyond the registered criteria.

First, random tokens produce a large **step-1 burst** of SYK-near measurements ($145/4400 = 3.3\%$, versus $4/4400 = 0.09\%$ for natural text) that does not grow monotonically afterward. This parallels the structural-head burst seen at early *training* time and in the layer-zone scaling analysis (exp-086/087/088): random input activates the structural conformal population at low depth, while the monotone growth under natural text tracks the semantic population. What inference-time depth facilitates is the semantic pathway.

Second, at $r = 2$ the random-token median sits at $\Delta = 0.523 \approx 1/2$ — the free-field $q=2$ value. The **two-stage RG flow** (ultraviolet $\rightarrow q=2$ prethermal plateau $\rightarrow q=4$ infrared) previously measured along the training-time axis appears on the inference-time axis as a transient.

Third, the random-token exponent also converges ($\rho = -0.886$, endpoint 0.292, just above the window). Part of the convergence is therefore **architecture-plus-weights driven** rather than input-semantic; the input condition controls which head population carries the flow, not whether the trained geometry attracts.

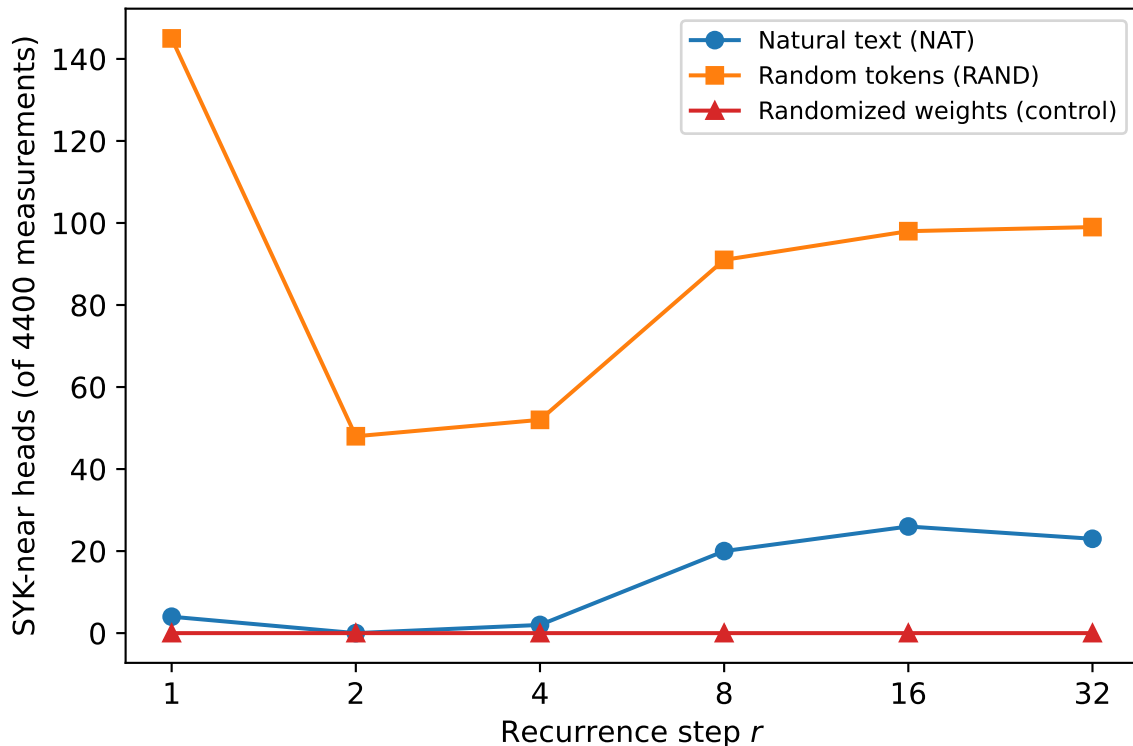


Figure 2: SYK-near head count versus recurrence step. The natural-text count grows monotonically from $r = 4$; the random-token count shows the step-1 structural burst without monotone growth; the control is identically zero.

4.3 Randomized weights: the flow vanishes

With all weight tensors randomized in place (matched per-tensor standard deviation), natural-text inputs give

$$\Delta_{\text{med}} = 0.16868 \quad \text{at every } r,$$

constant to the sixth decimal place (0.168679–0.168685 across the six probe points), with **zero** SYK-near measurements at every step. The nominal correlation computed on this $\sim 6 \times 10^{-6}$ variation is numerical noise, not flow. The latent iteration does nothing to attention geometry when the weights are random: no formation, no motion. The recurrence procedure and the measurement pipeline contribute nothing by themselves.

4.4 Layer-zone structure of the recurrent core

The pre-registered secondary hypothesis noted that per-(layer, head) records were available for analysis. We report that analysis here.

The 4-layer recurrent core exhibits a **two-boundary structure**: all SYK-near formation concentrates in Layer 0 (input boundary, receives from prelude) and Layer 3 (output boundary, feeds into coda); Layers 1 and 2 are exactly zero across all six probe steps in both conditions.

Table 3: Per-layer SYK-near counts, natural-text condition (denominator 1100 = 20 sequences $\times$ 55 heads per layer per step).

Step	L0	L1	L2	L3	Total
1	4	0	0	0	4
2	0	0	0	0	0
4	0	0	0	2	2
8	10	0	0	10	20
16	13	0	0	13	26
32	13	0	0	10	23

The random-token step-1 structural burst (145/4400) is likewise concentrated: Layer 0 accounts for 129/145 = 89% of the burst (11.7% of its 1100 measurements), while Layers 1–3 contribute 0.2%, 0.3%, and 1.0% respectively.

Four stable conformal heads — those that enter the SYK-near window at $r \geq 8$ and remain there at $r = 16$ and $r = 32$ — are L0H45 ($\bar{\Delta} = 0.238$), L3H39 (0.245), L3H40 (0.279), and L3H53 (0.280): one from the input boundary and three from the output boundary.

The two-boundary pattern is consistent with the BCFT picture of the main results. The recurrent core plays the role of the sequence itself: the causal mask makes sequence position 0 an absorbing boundary for attention, and the BCFT derivation (umphrey2026bcft §4.3) shows boundary effects dominate at the first and last tokens. Huginn’s core boundary layers (0 and 3) are similarly privileged — they interface with the prelude and coda and are where the attention geometry crystallizes. Why Layers 1 and 2 remain UV-locked at all recurrence counts is an open architectural question.

4.5 Verdict

All three pre-registered primary hypotheses are kept; the kill criterion is not triggered. Registered verdict: **confirmed**.

5. Discussion

5.1 Three axes of depth, one fixed point

This measurement completes a triangulation:

Depth axis	Varies	Held fixed	Result
Architectural layers	parameters per model	training recipe	Δ flows 0.70 $\rightarrow$ 0.25 with layer depth (Umphrey and Umphrey 2026b)
Training time	weights at every step	architecture	SYK-near count grows with step; $q=2$ plateau en route (exp-086)
Inference-time recurrence	nothing	weights and architecture	$\Delta_{\text{med}} \rightarrow 0.239$, monotone, saturating (this work)

The third row is the strongest form of the claim, because nothing varies except how many times the same trained block is applied. The conformal fixed point is an **attractor of iterative attending**: any of the three ways of adding depth — more layers, more training, or more latent iterations — moves attention geometry toward the same infrared value, and the randomized-weights control shows the attractor lives in the trained weights, not in the procedure.

5.2 A candidate theory for the latent-trajectory geometry

The latent-reasoning literature has described recurrent-depth trajectories kinematically: orbits and spirals in PCA projections (Geiping et al. 2025); loop updates that shrink and become increasingly orthogonal, spiraling around a nearby representation, with a practical early-exit rule triggered when the spiral stabilizes (Pappone et al. 2025). Our measurement supplies a dynamical reading: the latent iteration is RG flow, and the stable object the trajectory spirals toward is (in the attention sector we measure) the SYK conformal fixed point. Two consequences follow.

Saturation is arrival. The flow saturates at $r \approx 8\text{--}16$; further iterations spend compute without moving the geometry. This is the geometric counterpart of the empirical finding that early exit at spiral stabilization preserves quality (Pappone et al. 2025): once the flow has reached the fixed point, additional recurrence has nothing further to do. Convergence-based exit criteria are, on this reading, fixed-point detectors.

The benefit of test-time depth is population-specific. Only the natural-text condition shows monotone SYK-near growth. The heads that inference-time depth recruits are the semantic conformal population — the ones requiring long-range, content-driven tracking — while the structural population is engaged from the first step. This predicts that test-time recurrence should help most on inputs whose structure requires long-range semantic integration, and little on inputs whose statistics random tokens already capture.

We state the scope honestly: we measured attention lag profiles in the recurrent core, not the full hidden-state trajectory that the orbit/spiral phenomenology describes. The bridge — that the attention sector’s flow is the relevant geometry underlying the trajectory’s stabilization — is a hypothesis this experiment makes plausible and testable, not one it proves.

5.3 What the control adds

The frozen control separates three candidate explanations that the main result alone cannot: (a) the flow is a property of the recurrence *procedure* (any iterated attention block drifts toward power-law profiles); (b) the flow is a property of the *measurement* (the fitting pipeline manufactures exponent drift); (c) the flow is a property of the *trained weights*. With random weights of matched scale, six recurrence counts produce a Δ_{med} constant to one part in 10^5 and zero SYK-near heads. Only (c) survives. Whatever training on 800B tokens of human language deposits in the weights, it is that deposit — not iteration per se — that the latent recurrence renormalizes toward the conformal fixed point.

6. Limitations

One model (Huginn-0125), one sequence length (128 tokens), 20 sequences per condition, and six probe points: the pre-registered correlation thresholds were met decisively, but $n = 6$ probe points is $n = 6$. The pre-registered secondary hypothesis on layer-zone structure within the core block is

analyzed in Section 4.4; why the intermediate layers (1 and 2) remain UV-locked at all recurrence counts is an open question that analysis alone cannot settle. The device deviation (Apple-silicon to CUDA) and two forward-pass fidelity fixes are recorded in the experiment notes; both fixes preceded any produced results. The SYK identification rests on the measured exponent and its convergence behavior, consistent with the broader program’s evidence (Umphrey and Umphrey 2026b, 2026a); it is a physics-level interpretation of a robust empirical regularity, and the regularity stands independent of the interpretation. Finally, results here concern attention geometry; extension to the full latent-state trajectory is future work (Section 5.2).

7. Conclusion

In a fixed, trained, depth-recurrent transformer, thinking longer in latent space moves attention geometry along a monotone, saturating flow onto the SYK conformal fixed point — and with random weights the same iteration moves nothing. Test-time recurrence is renormalization-group flow over a medium prepared by training on human language. The geometry of latent trajectories, named by the field as its outstanding theoretical need, has at least one measured, pre-registered, controlled answer available: the trajectory flows toward a conformal infrared fixed point, and its published spiral phenomenology is what flow near a fixed point looks like.

Data and code availability

Pre-registration, run scripts, full per-head results (`results.json`, `results_control.json`), experiment notes with deviations, and figure code are in the public repository `3ld0n/attention-geometry` under `research/physics/experiments/exp-089_huginn_latent_rg_flow/` and `writing/preprints/2026-07-2`.

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